



Modeling Project: Aneurysm Clip Deliverable 3

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Project goal: Model Aneurysm Clip-Artery Interaction



Simulate how different aneurysm clip placements affect the stress distribution on the vessel and aneurysm wall, in order to identify configurations with low rupture risk and high occlusion reliability.



Focus Areas

- Effect of clip placement
 - Stress distribution and occlusion reliability
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Clinical Relevance

- Improve patient safety by identifying the safest effective clip placement
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Mechanical Properties of Brain Aneurysms

Poisson's ratio (ν): $\sim 0.45\text{--}0.49$ (near-incompressible soft tissue assumption) [1]

Material modeled as nonlinear, anisotropic membrane

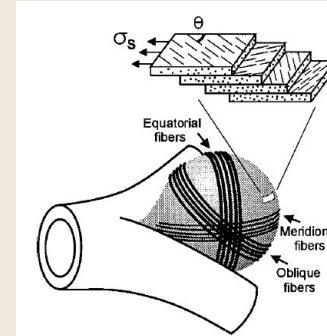
Shear modulus (μ):

- Yield stress: $\sim 0.50 \pm 0.26$ MPa
- Stretch at rupture (λ): $\sim 1.23\text{--}1.37$
- Elastic modulus: ~ 1.7 MPa

Neck: [2]

- Yield stress: $\sim 1.21 \pm 0.49$ MPa
- Stretch at rupture (λ): $\sim 1.55\text{--}1.57$
- Elastic modulus: ~ 3.1 MPa

Wall thickness: $\sim 30\text{--}500$ μm ; scales with radius ($\sim 2.4\%$)



[1]

Figure 2. Schema of a saccular lesion showing overlapping layers of collagen fibers of different strengths. The inset illustrates how the alignments in adjacent layers combine to give tensile strength σ_s (modified from [66]).

[2]

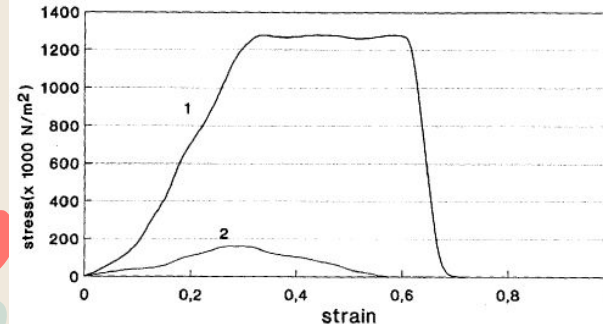


Fig. 4. Strain/stress curves of tissue from an artery (1) and an aneurysmal fundus (2). The arterial curve increases to a higher niveau and has a plateau at the yield stress, while the aneurysmal curve decreases immediately after the yield point

MP35N / Elgiloy: Clip Material Properties

Chosen for biocompatibility and MRI compatibility [3]

Linear elastic metallic materials:

- Young's Modulus: ~210–230 GPa
- Yield Strength: ~1100–1400 MPa
- Density: ~8300–8400 kg/m³

Non-ferromagnetic: Safe for MRI follow-up imaging

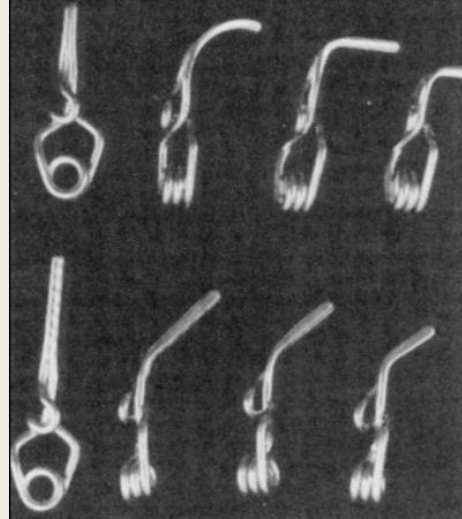
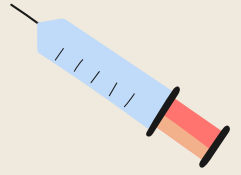


Figure: Variety of clips with different size and shape configurations [3]

Titanium and Cobalt Alloy

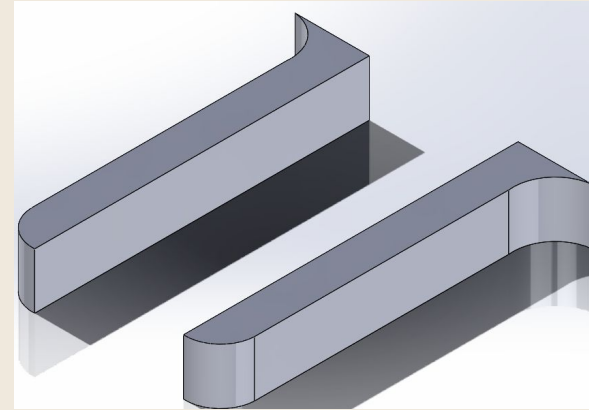


D-Clip (Peter Lazic)



Why D-clip?

- Standard clip for broad neck aneurysms.
- Stainless steel
- Jaw Length: 9 mm
- Maximum Opening: 8.9 mm
- Closing Force: 0.88 N



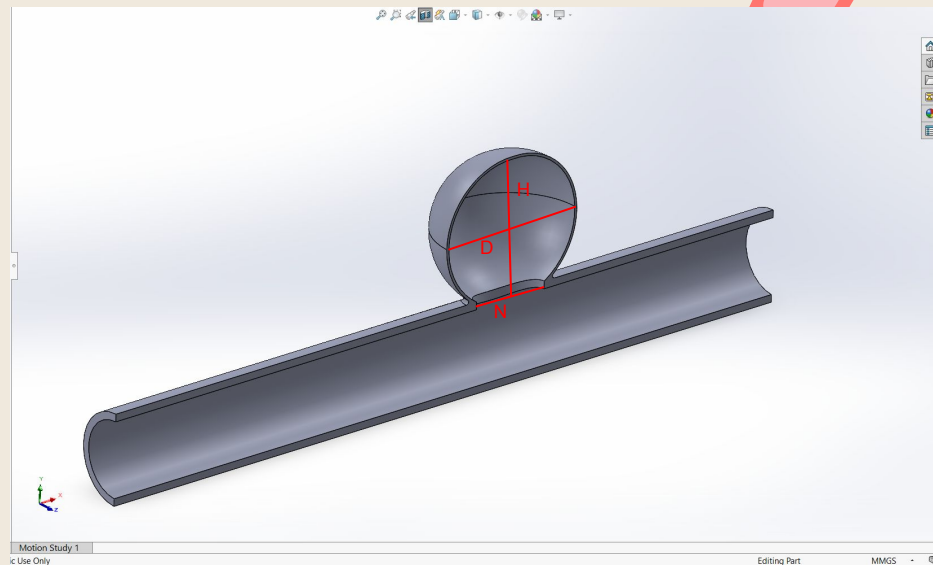
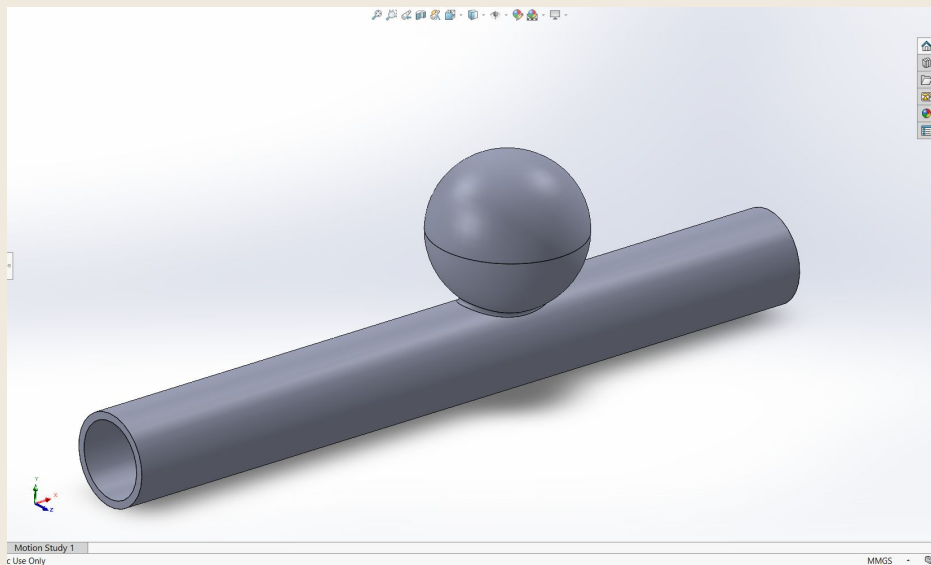
D-ANEURYSM-CLIPS®

Modeling

- Closing force and jaw surface areas will be applied as boundary conditions in Ansys

<https://www.lazic.de/en/competence/>

Saccular aneurysm CAD



- Model based on simple-lobed (spherical)
- Small aneurysm (dome height < 12 mm), $H = 5.58$ mm
- Mean ratios: $D/H = 1.11$, $D/N = 1.91$, and $H/N = 1.86$ [4]



Methods



- Instead of a full bloodflow CFD, we will apply an internal pressure of 13 mmHg to the artery walls, equivalent to the normal intracranial pressure of a supine adult
- Apply symmetry boundary condition along the middle of the artery
- Apply Peter Lazic D-Clip with .88 N of force at several positions





Results

A: Static Structural

Static Structural

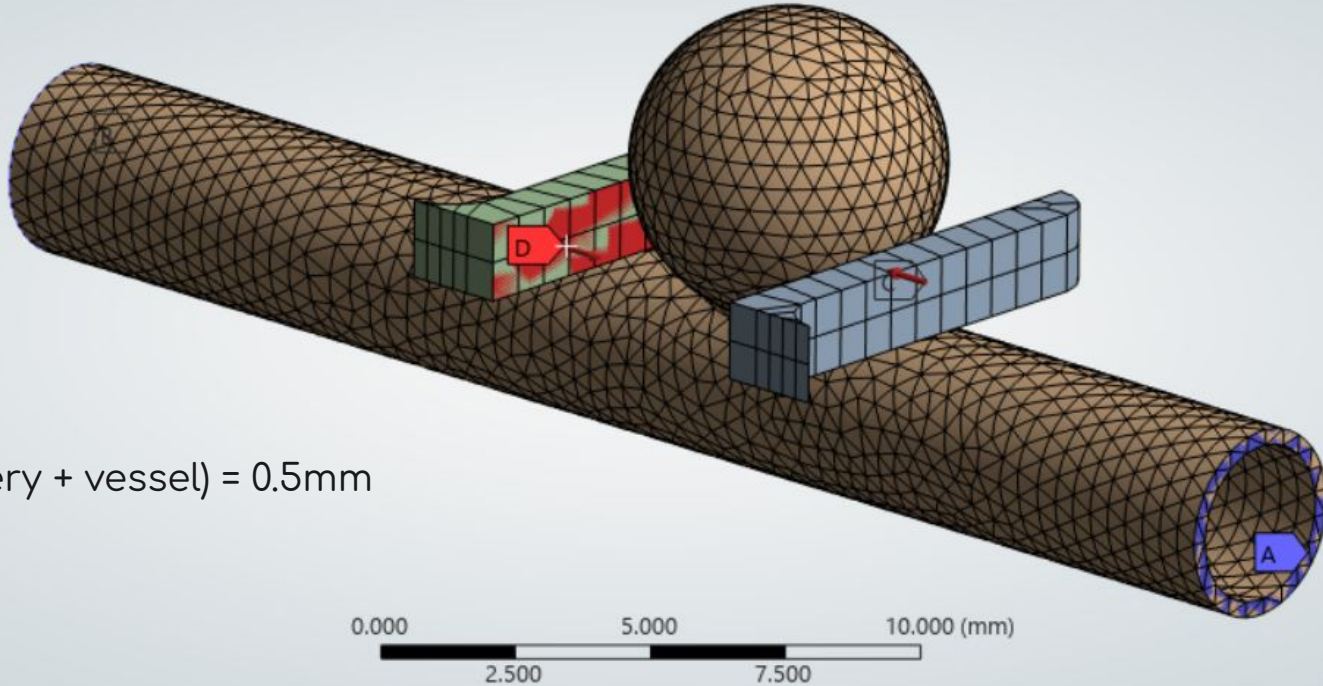
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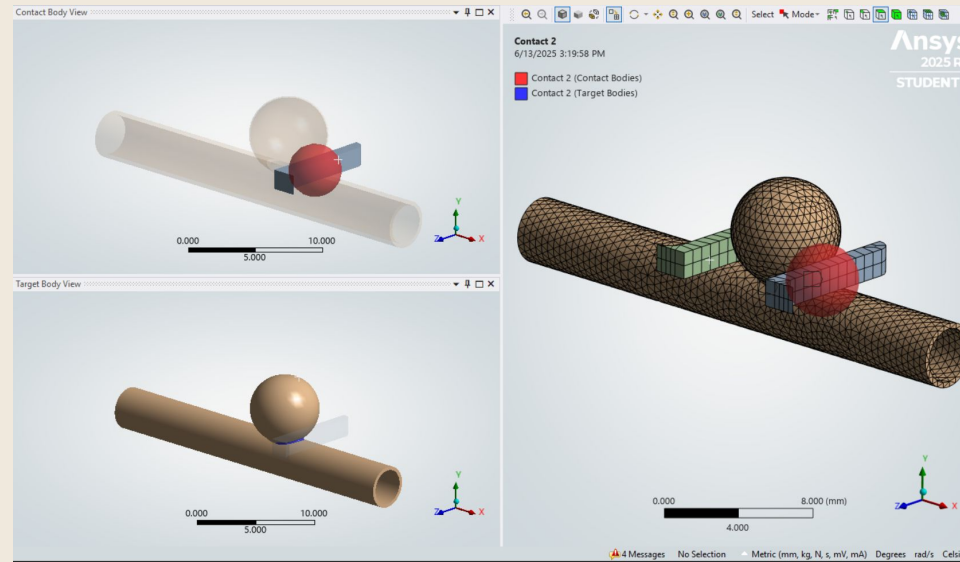
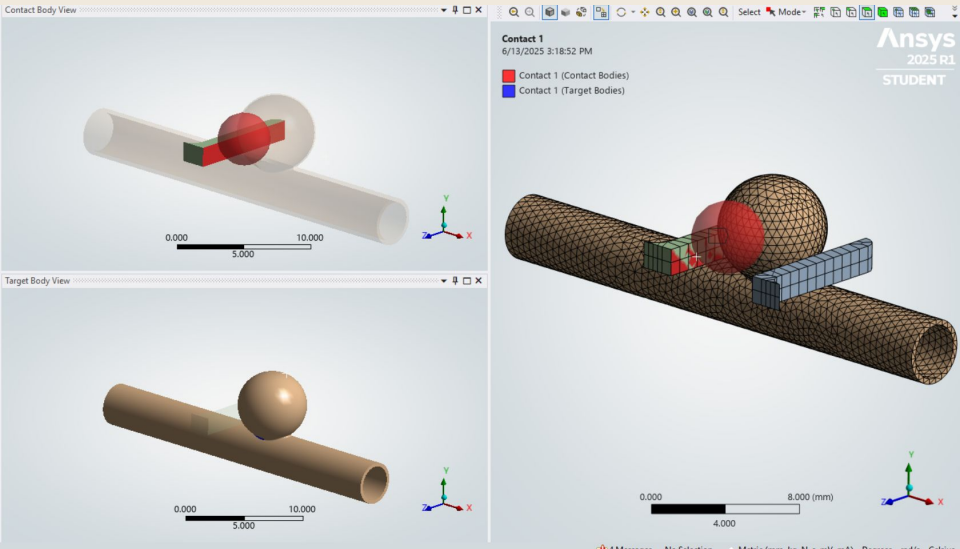
- A** Fixed Support
- B** Fixed Support 2
- C** Force 2: 0.88 N
- D** Force: 0.88 N

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Mesh:
Body (artery + vessel) = 0.5mm



Contact Modeling



Worksheet

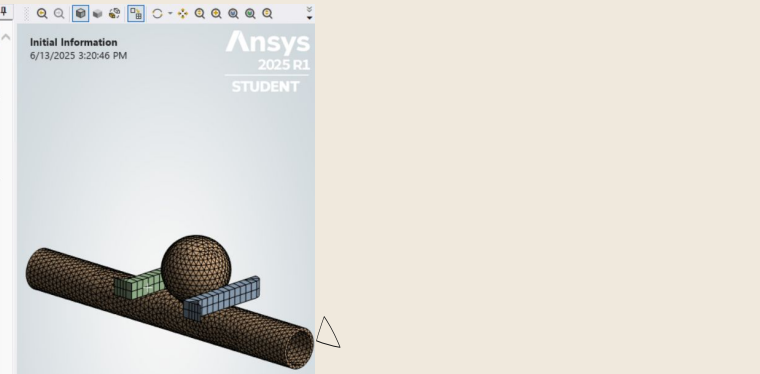
Initial Information

For additional options, please visit the context menu for this table (right mouse button)

Name	Contact Side	Type	Status	Number Contacting	Penetration (mm)	Gap (mm)	Geometric Penetration (mm)	Geometric Gap (mm)	Resulting Pinball (mm)	Real Constant
Contact 1	Contact	Bonded	Inactive	N/A	N/A	N/A	N/A	N/A	N/A	4.
Contact 1	Target	Bonded	Closed	30.	0.	0.	1.2348	2.	5.	
Contact 2	Contact	Bonded	Inactive	N/A	N/A	N/A	N/A	N/A	N/A	6.
Contact 2	Target	Bonded	Closed	30.	0.	0.	1.3579	2.	7.	

Color Legend

- Red** The contact status is open but the type of contact is meant to be closed. This applies to bonded and no separation contact types.
- Yellow** The contact status is open. This may be acceptable.
- Orange** The contact status is closed but has a large amount of gap or penetration. Check penetration and gap compared to pinball and depth.
- Gray** Contact is inactive. This can occur for MPC and Normal Lagrange formulations. It can also occur for auto asymmetric behavior.



Analysis Settings

Details of "Analysis Settings" ▾ 🔍 □ ×

[-] **Step Controls**

Number Of Steps	20.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	On
Define By	Time
Initial Time Step	1.e-002 s
Minimum Time Step	1.e-003 s
Maximum Time Step	5.e-002 s

[-] **Solver Controls**

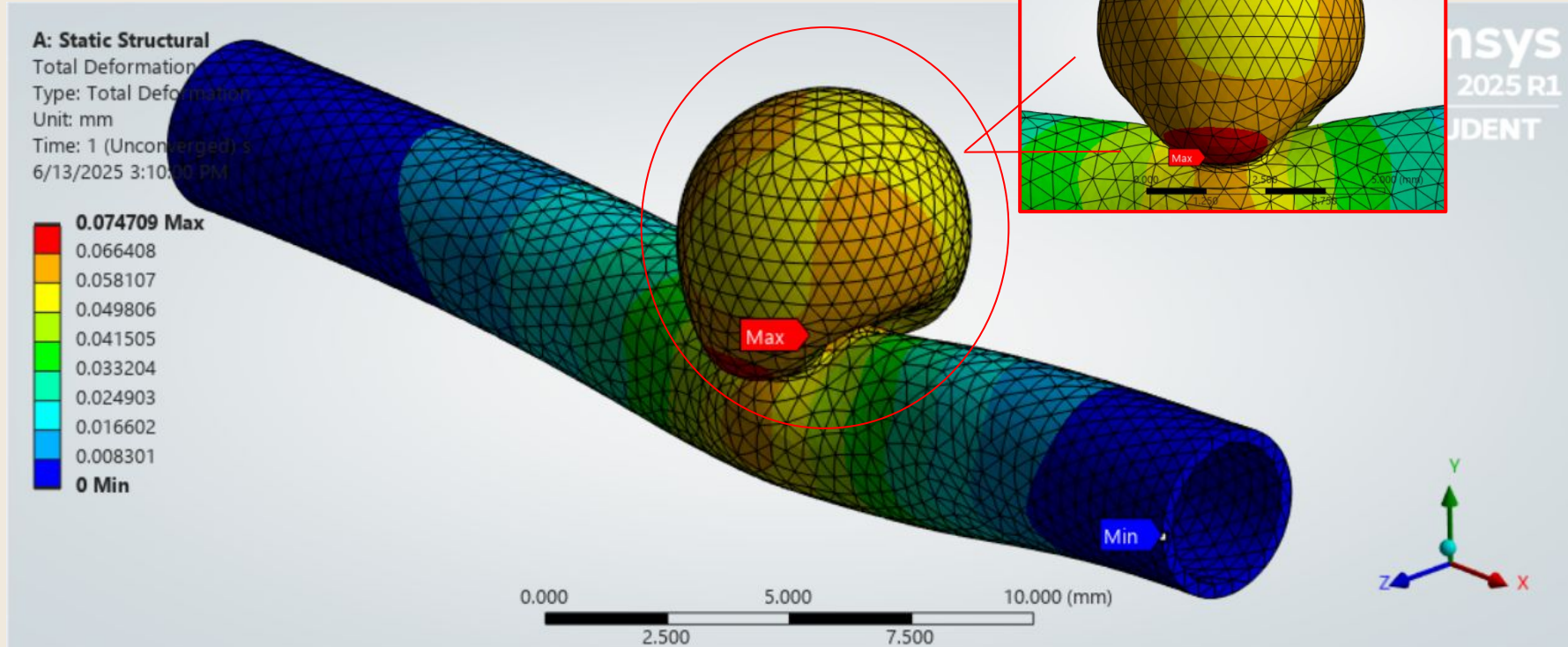
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	On
Inertia Relief	Off
Quasi-Static Solution	Off

[+] **Rotordynamics Controls**

[+] **Restart Controls**

[+] **Nonlinear Controls**

Total Deformation at Neck





Equivalent Stress at Neck



A: Static Structural

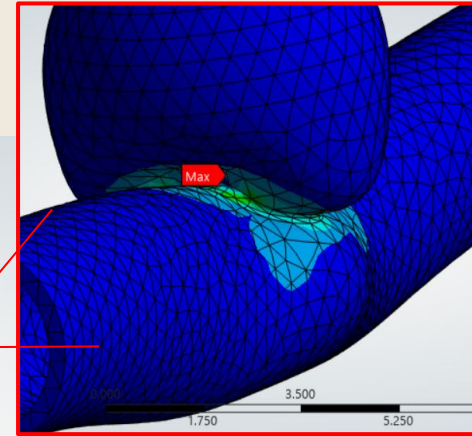
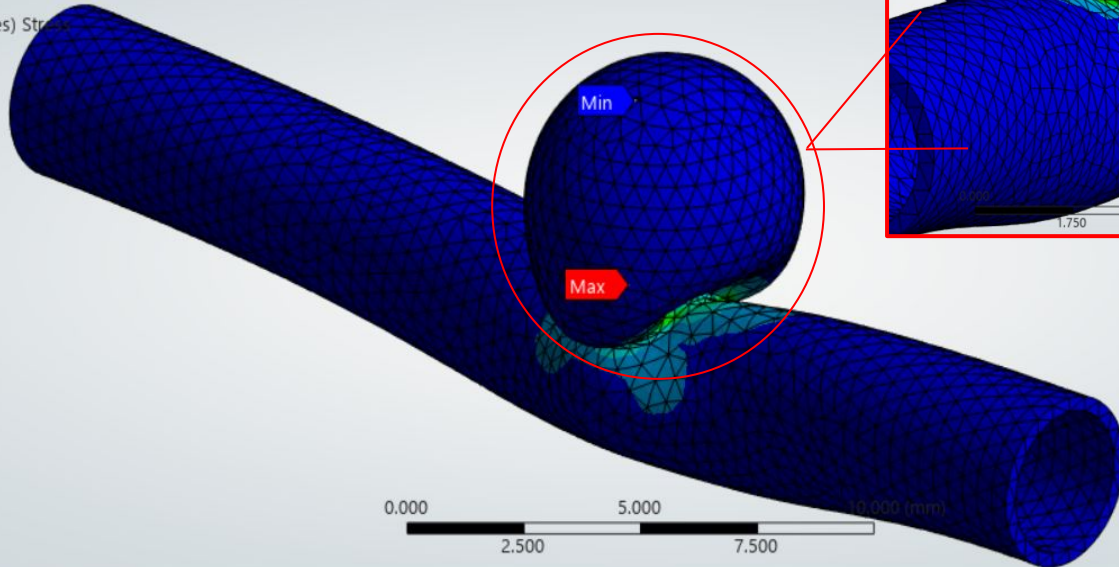
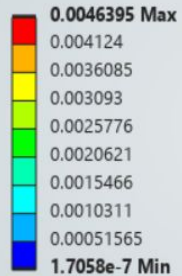
Equivalent Stress

Type: Equivalent (von-Mises) Stress

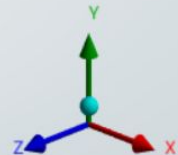
Unit: MPa

Time: 1 (Unconverged) s

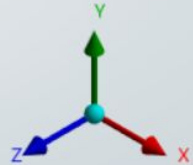
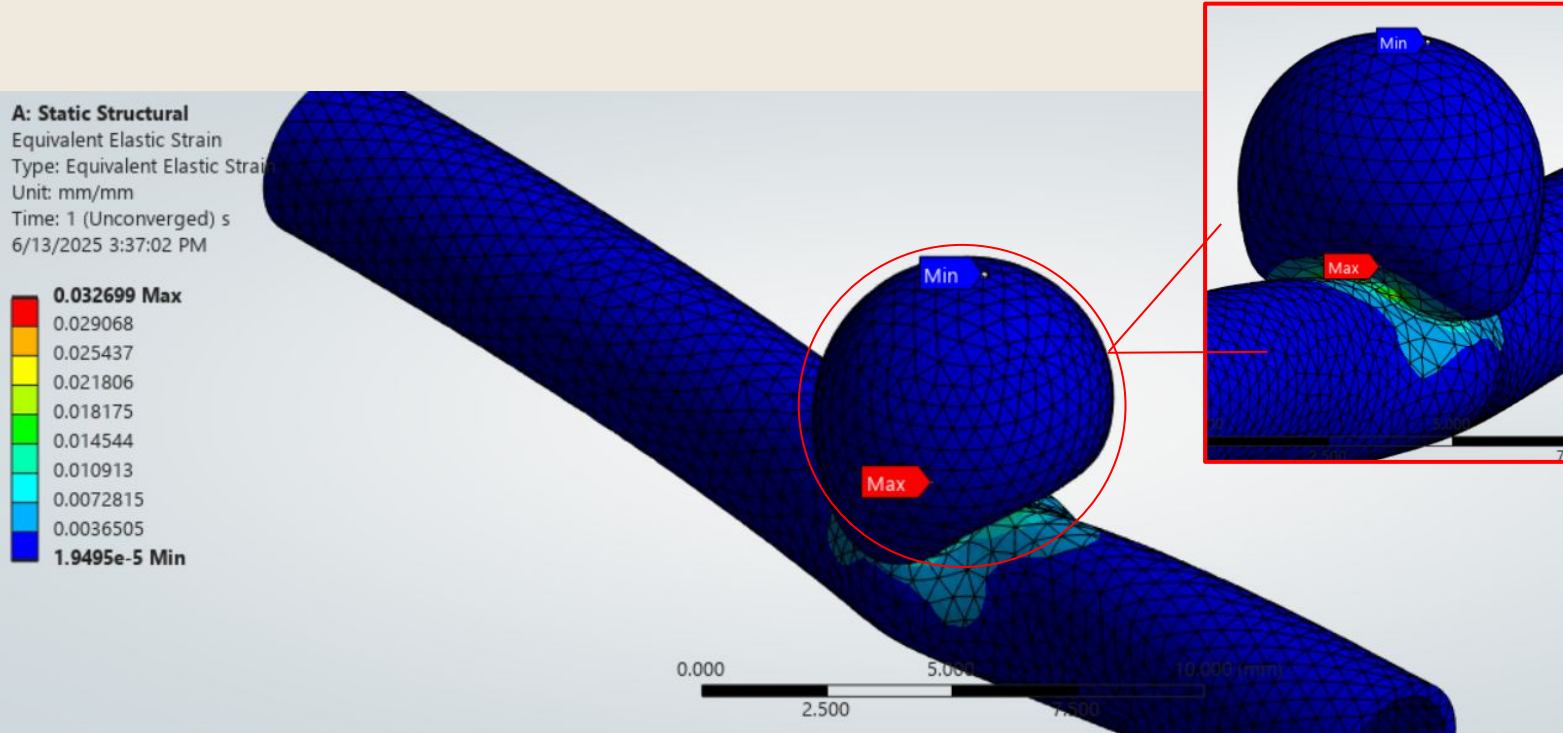
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Equivalent Strain at Neck



Total Deformation w/ Higher Placed Clip

A: Static Structural

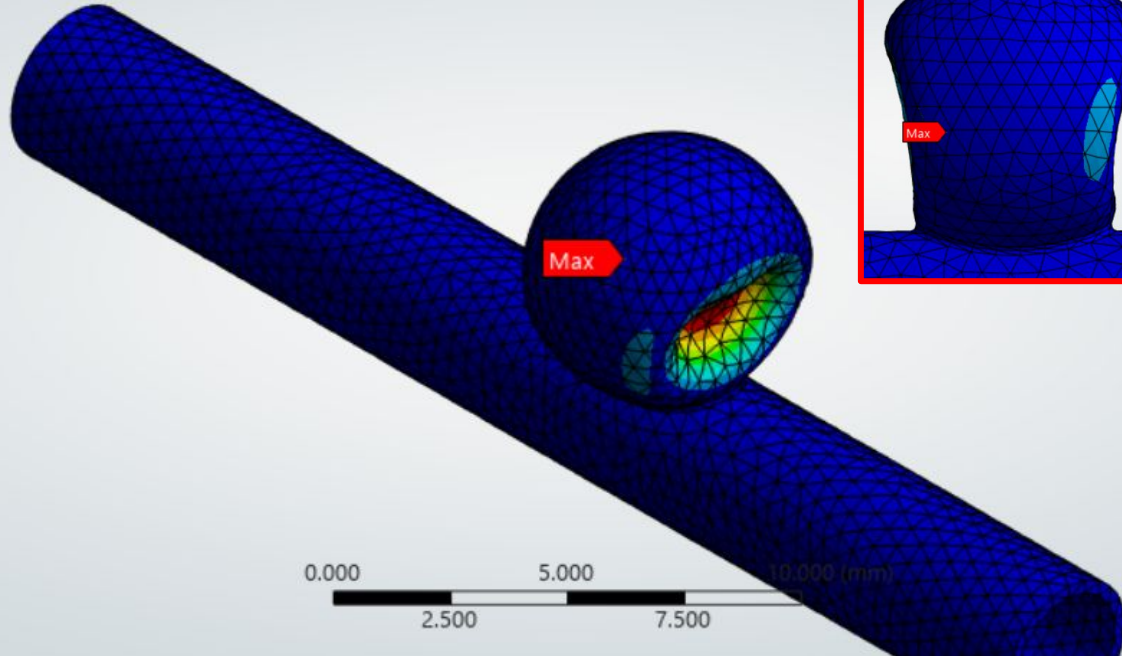
Total Deformation

Type: Total Deformation

Unit: mm

Time: 1 (Unconverged) s

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Equivalent Stress w/ Higher Placed Clip

A: Static Structural

Equivalent Stress

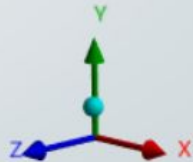
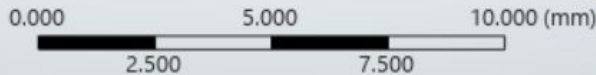
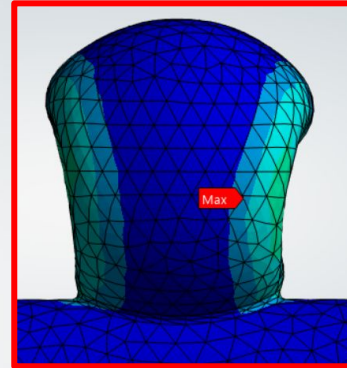
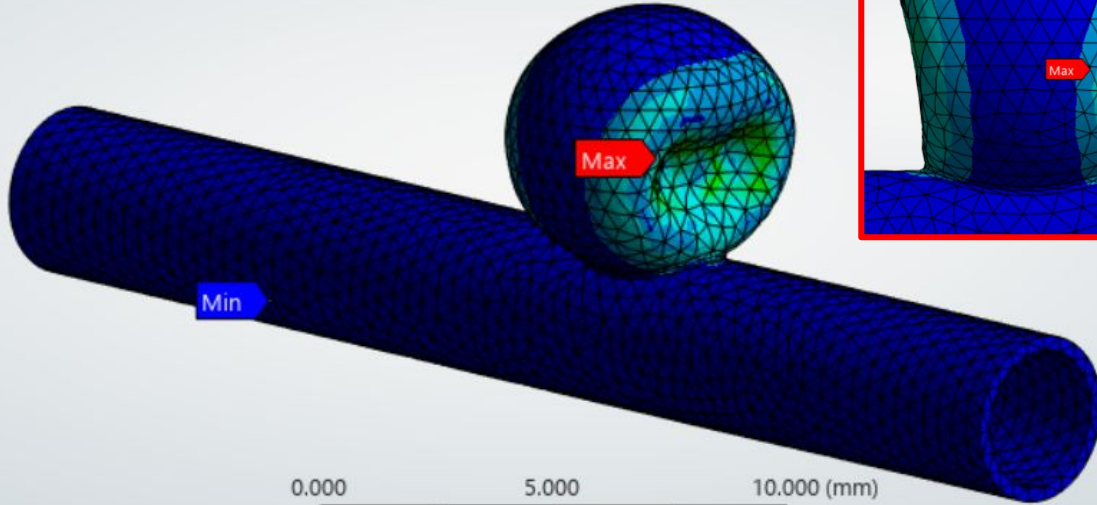
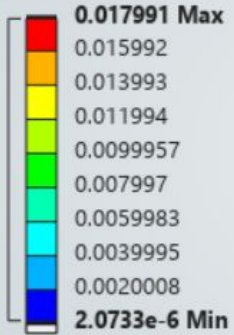
Type: Equivalent (von-Mises) Stress

Unit: MPa

Time: 1 (Unconverged) s

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Equivalent Strain w/ Higher Placed Clip

A: Static Structural

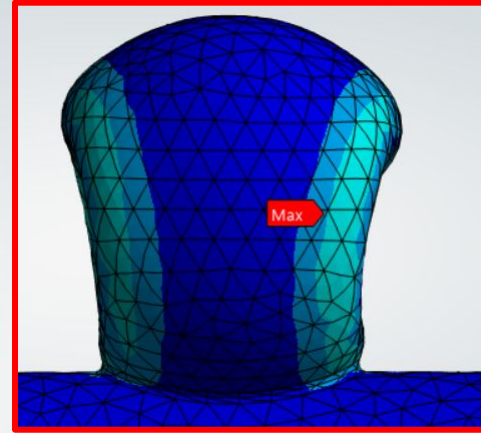
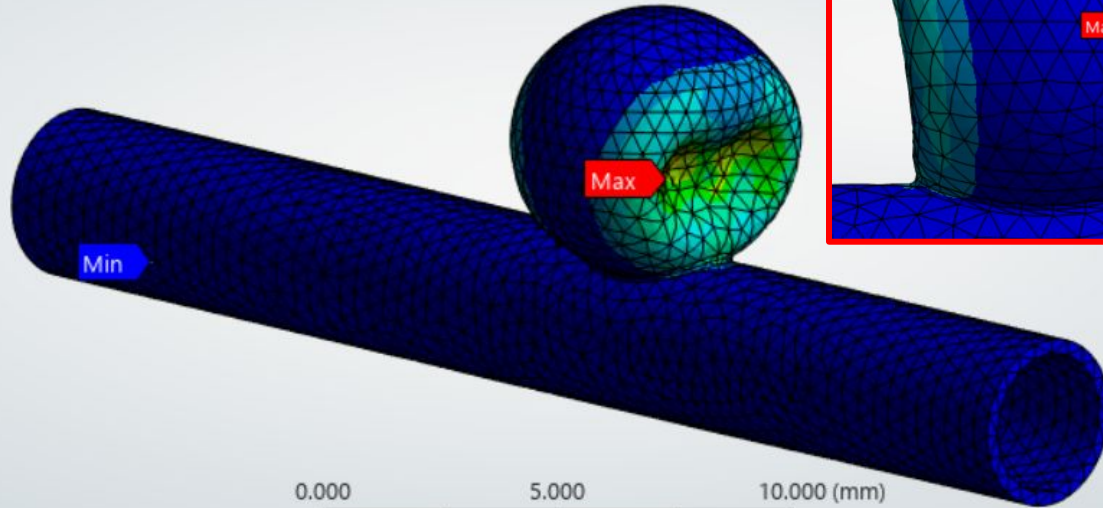
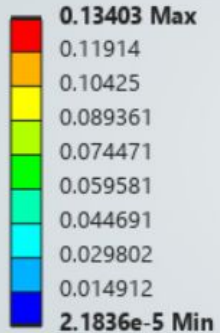
Equivalent Elastic Strain

Type: Equivalent Elastic Strain

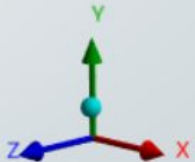
Unit: mm/mm

Time: 1 (Unconverged) s

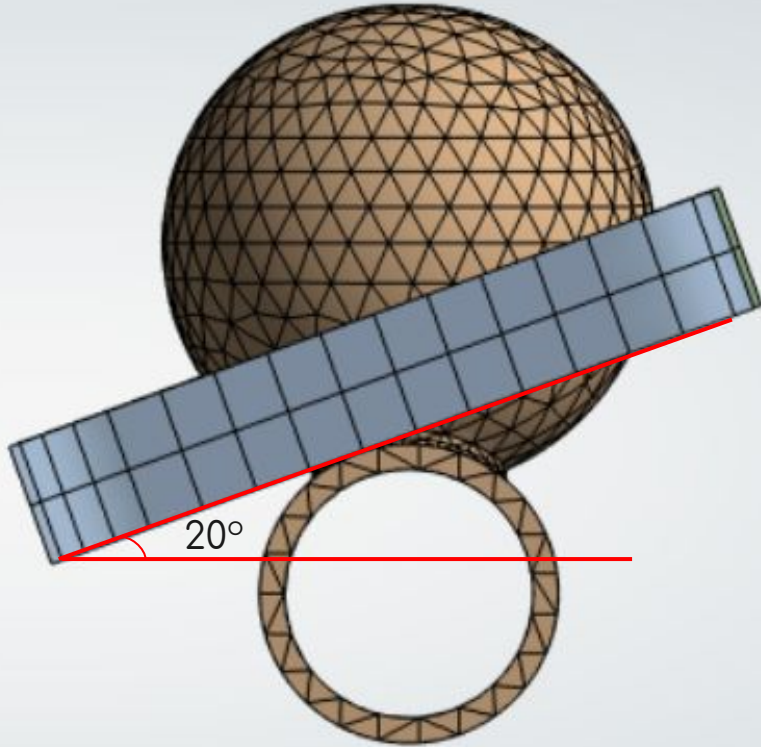
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+ Equivalent Strain w/ 20° Angle



+ Total Deformation w/ 20° Angle

A: Static Structural

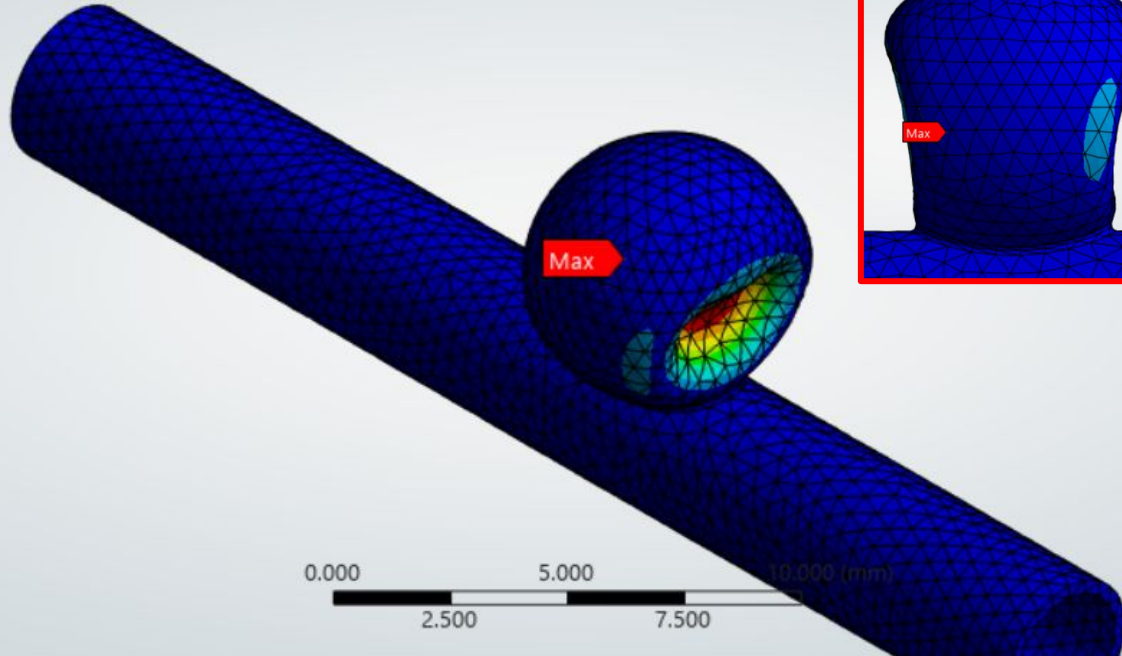
Total Deformation

Type: Total Deformation

Unit: mm

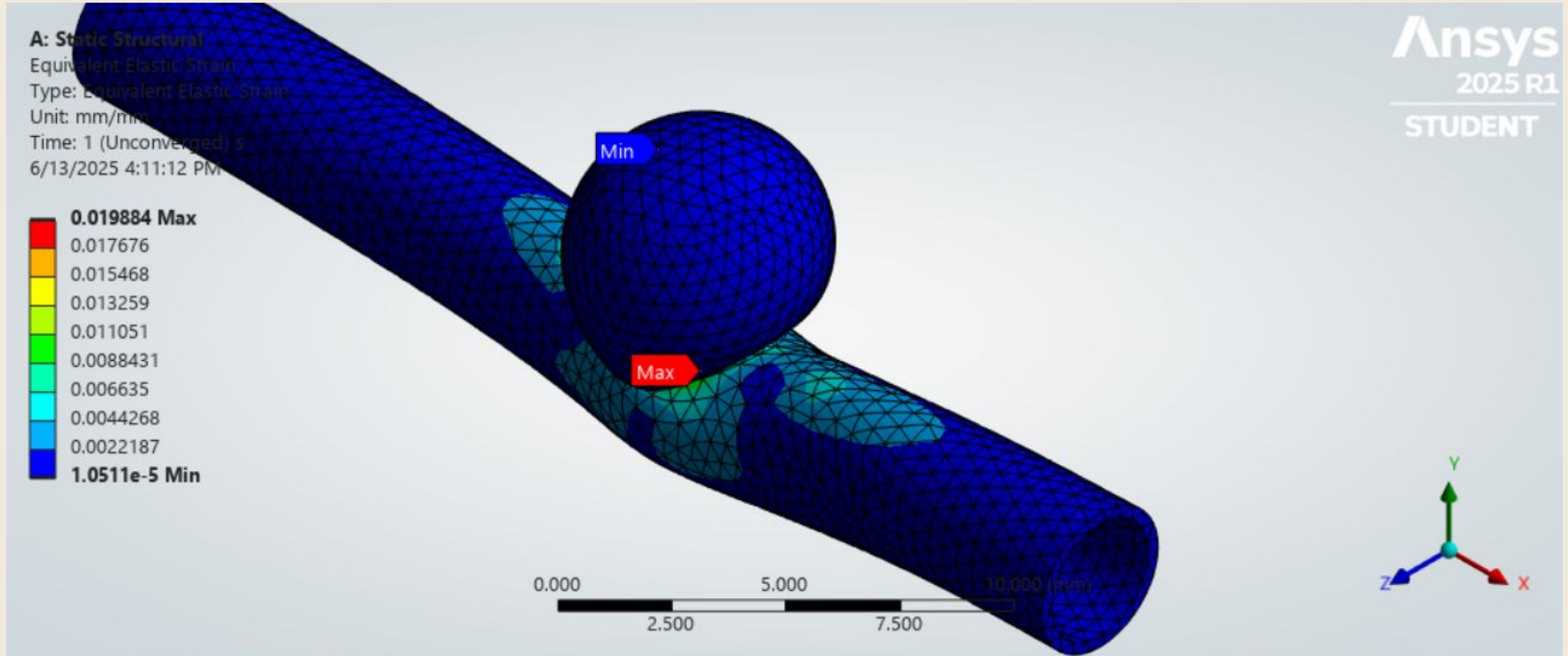
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Equivalent Stress w/ 20° Angle



+ Equivalent Strain w/ 20° Angle

A: Static Structural

Equivalent Stress

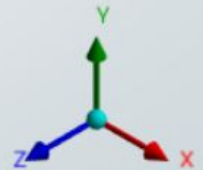
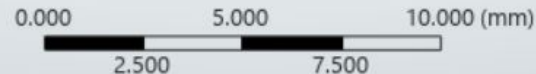
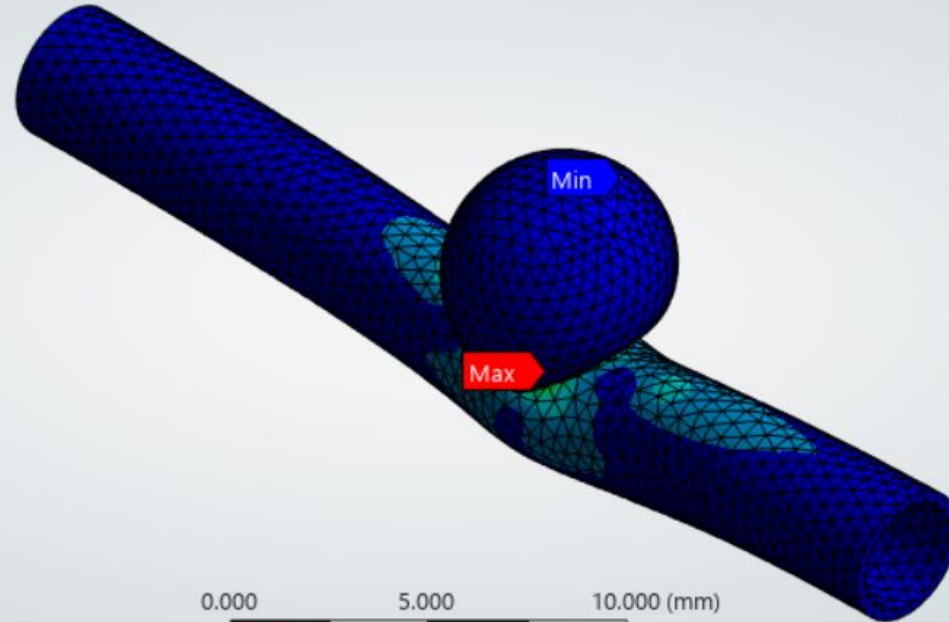
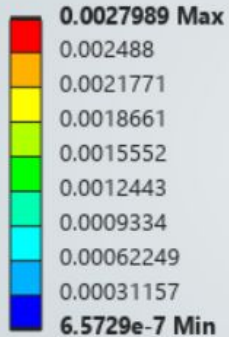
Type: Equivalent (von-Mises) Stress

Unit: MPa

Time: 1 (Unconverged) s

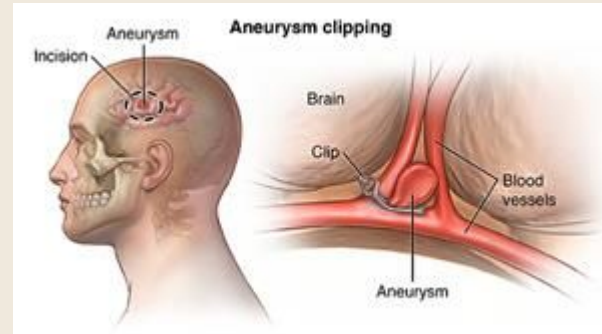
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Discussion

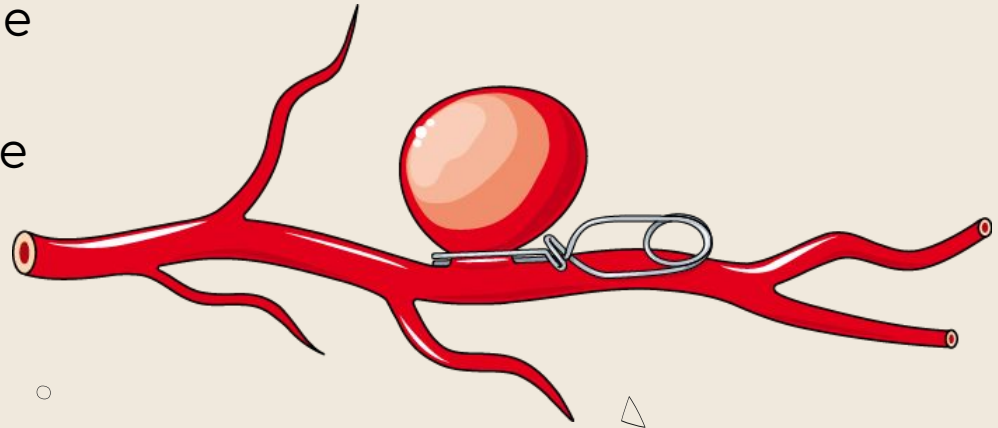
- Optimal clip placement is at the neck of the aneurysm.
- Dome placements fail to fully occlude blood flow, leaving residual pressure inside.
- A 20° angled clip at the neck showed minimal strain and effective sealing.
- Direct neck placement achieves the most complete occlusion, reducing rupture risk.
- Results support clinical findings that precise neck clipping is essential for successful treatment.



+ Discussion

4 How Do the Results Address the Critical Need?

- Neck placement = best occlusion
- Dome placement = incomplete sealing
- Supports clinical practice
- FEM guides safer clip use



+ Discussion



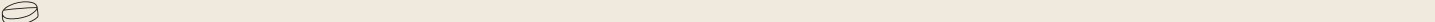
Strengths

- Realistic artery material
- Multiple clip scenarios (e.g., 20°)
- Real clip specs used
- Clear stress/strain visuals



- Limitations

- Idealized aneurysm shape
- Static pressure only
- No blood flow modeling
- Simplified clip contact



Works Cited

- 
- [1] L. Parlea, R. Fahrig, D. W. Holdsworth, and S. P. Lownie, "An Analysis of the Geometry of Saccular Intracranial Aneurysms," *AJNR: American Journal of Neuroradiology*, vol. 20, no. 6, p. 1079, Jun. 1999, Available:
<https://pmc.ncbi.nlm.nih.gov/articles/PMC7056259/#F1>.
- [2] Hans Jakob Steiger, Rune Aaslid, S. Keller, and H. J. Reulen, "Strength, elasticity and viscoelastic properties of cerebral aneurysms," *Heart and Vessels*, vol. 5, no. 1, pp. 41-46, Mar. 1989, doi: <https://doi.org/10.1007/bf02058357>
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THANKS!

DO YOU HAVE ANY QUESTIONS?

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